

New Trade Theory II

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The **Melitz model (2003)** is the **most influential extension** of New Trade Theory. It introduces **firm heterogeneity** in **productivity** and explains why only some firms export, why exporters are larger and more productive, and how trade causes important reallocation effects within industries.

Core Idea: Within this framework, the **self-selection effect** is one of the most important and empirically robust concepts. **Only the most productive firms become exporters**

1. Key Assumptions

- Monopolistic competition + differentiated varieties (like Krugman)
- Firms differ in **productivity** (ϕ) – drawn from a Pareto distribution
- Increasing returns to scale (fixed cost + constant marginal cost)
- Two types of fixed costs:
 - Domestic fixed cost: f_d and Export fixed cost: f_x (higher than f_d)
- Iceberg trade cost: $\tau > 1$ (to deliver 1 unit abroad, τ units must be shipped)
- CES preferences with elasticity of substitution $\sigma > 1$

2. Core Structure of the Model

Firm Revenue and Profit

A firm with productivity ϕ has:

- **Marginal cost:** $\frac{w}{\phi}$, **Domestic price:** $p_d(\phi) = \left(\frac{\sigma}{\sigma-1}\right) \frac{w}{\phi}$
- **Export price** (in foreign market): $p_x(\phi) = \left(\frac{\sigma}{\sigma-1}\right) \frac{\tau w}{\phi}$

Revenue:

$$r_d(\phi) = R \left(\frac{p_d(\phi)}{P} \right)^{1-\sigma} \quad r_x(\phi) = R^* \left(\frac{p_x(\phi)}{P^*} \right)^{1-\sigma}$$

Where R = total domestic expenditure, P = price index.

then **profit**

$$\pi_d(\phi) = \frac{r_d(\phi)}{\sigma} - wf_d, \quad \pi_x(\phi) = \frac{r_x(\phi)}{\sigma} - wf_x$$

3. Key Cutoffs (Most Important Equations)

Firms decide whether to produce and export based on **productivity**:

- **Domestic production cutoff** (ϕ_d^*): **Minimum productivity** to serve the domestic market
 $\pi_d(\phi_d^*) = 0 \Rightarrow r_d(\phi_d^*) = \sigma w f_d$
- **Export cutoff** (ϕ_x^*): **Minimum productivity** to export $\pi_x(\phi_x^*) = 0 \Rightarrow r_x(\phi_x^*) = \sigma w f_x$

Moreover, their profit is

$$\bullet \pi_d(\phi) = w f_d \left[\left(\frac{\phi}{\phi_d^*} \right)^{\sigma-1} - 1 \right] \text{ and } \pi_x(\phi) = w f_x \left[\left(\frac{\phi}{\phi_x^*} \right)^{\sigma-1} - 1 \right]$$

Because of **trade costs** (τ) and **higher fixed export cost** (f_x), we get:

$$\phi_x^* > \phi_d^*$$

Only the most productive firms export.

Summary of Key Results

1. **Revenue increases with productivity:** $r(\phi) \propto \phi^{\sigma-1}$. More productive firms sell more because they charge lower prices.
2. **Profit is convex in productivity:** $\pi(\phi) \propto \phi^{\sigma-1}$ **constant**. Small differences in productivity lead to large differences in profit.
3. **Selection effect:** Only firms with $\phi \geq \phi_d^*$ survive domestically. Only firms with $\phi \geq \phi_x^*$ export. Since $\phi_x^* > \phi_d^*$, the most productive firms export, while less productive firms serve only the domestic market, and the least productive firms exit.
4. **Reallocation effect:** Trade liberalization (lower τ) raises ϕ_d^* , forcing the least productive firms to exit. Resources are reallocated toward more productive firms, raising aggregate productivity. This is the core mechanism through which trade generates gains in the Melitz model.

4. Productivity Distribution (Pareto)

Productivity ϕ is drawn from a Pareto distribution:

$$G(\phi) = 1 - \left(\frac{\phi_{min}}{\phi} \right)^k \quad \text{for } \phi \geq \phi_{min}$$

Where $k > \sigma - 1$ (shape parameter). This assumption delivers very nice closed-form solutions.

5. Main Predictions of the Melitz Model

Prediction	Explanation
Selection into Exporting	Only firms with $\phi > \phi_x^*$ export
Exporter Premium	Exporters are larger, more productive, and pay higher wages
Intra-industry Reallocation	Trade liberalization → low-productivity firms exit, resources move to more productive firms
Productivity Gains	Aggregate industry productivity rises even without new technology
Gains from Trade	Larger than in Krugman (includes reallocation gains)

6. Gains from Trade in Melitz Model

Trade generates **three types of gains**:

1. **Variety gains** (same as Krugman)
2. **Pro-competitive gains** (lower markups due to more competition)
3. **Reallocation / Selection gains** — most important new channel

When trade opens:

- Domestic cutoff ϕ_d^* rises $\rightarrow$ least productive firms exit
- Export cutoff ϕ_x^* falls $\rightarrow$ more firms start exporting
- Resources reallocate toward higher-productivity firms $\rightarrow$ average industry productivity increases

This **reallocation effect** can be **very large**.

7. Key Equation: Aggregate Productivity

Average industry productivity ($\tilde{\phi}$) rises with trade openness. In equilibrium, the weighted average productivity is:

$$\tilde{\phi} = \left[\frac{1}{1 - G(\phi_d^*)} \int_{\phi_d^*}^{\infty} \phi^{\sigma-1} dG(\phi) \right]^{\frac{1}{\sigma-1}}$$

With Pareto distribution, this simplifies nicely and shows clear productivity gains from lower trade costs.

Mathematical Derivation

1. Firm Profit and Zero-Profit Conditions

Firm Revenue

A firm with productivity ϕ faces CES demand. Its revenue in the domestic market is:

$$r_d(\phi) = R \left(\frac{p_d(\phi)}{P} \right)^{1-\sigma}$$

where:

- R = total domestic expenditure (income)
- P = domestic price index; σ = elasticity of substitution

The optimal price (markup pricing):

$$p_d(\phi) = \left(\frac{\sigma}{\sigma - 1} \right) \frac{w}{\phi} = \mu \frac{w}{\phi}$$

Substitute price into revenue:

$$r_d(\phi) = R \left(\frac{\mu w / \phi}{P} \right)^{1-\sigma} = R \left(\frac{\mu w}{P} \right)^{1-\sigma} \phi^{\sigma-1}$$

Let $B = R \left(\frac{\mu w}{P} \right)^{1-\sigma}$ (a market demand index). Then:

$$r_d(\phi) = B \cdot \phi^{\sigma-1}$$

Export revenue (with iceberg trade cost τ):

$$r_x(\phi) = B^* \cdot (\phi/\tau)^{\sigma-1}$$

Operating Profit

Because of CES demand, operating profit (revenue minus variable cost) is revenue divided by σ :

$$\pi_d^{op}(\phi) = \frac{r_d(\phi)}{\sigma} = \frac{B}{\sigma} \phi^{\sigma-1}, \quad \pi_x^{op}(\phi) = \frac{r_x(\phi)}{\sigma} = \frac{B^*}{\sigma} \left(\frac{\phi}{\tau} \right)^{\sigma-1}$$

Zero-Profit Conditions (Net Profit = 0)

Domestic market:

$$\pi_d(\phi) = \frac{r_d(\phi)}{\sigma} - wf_d = 0, \quad \frac{B}{\sigma} \phi_d^{*(\sigma-1)} = wf_d$$

Export market:

$$\frac{B^*}{\sigma} \left(\frac{\phi_x^*}{\tau} \right)^{\sigma-1} = wf_x$$

Solving for the **productivity cutoffs**:

$$\phi_d^* = \left(\frac{\sigma w f_d}{B} \right)^{\frac{1}{\sigma-1}}, \quad \phi_x^* = \tau \left(\frac{\sigma w f_x}{B^*} \right)^{\frac{1}{\sigma-1}}$$

Key Relationship:

$$\phi_x^* > \phi_d^* \quad (\text{because } f_x > f_d \text{ and } \tau > 1)$$

- Only firms with productivity $\phi \geq \phi_d^*$ produce for the domestic market.
- Firms with $\phi \geq \phi_x^*$ also export.

2. Free Entry Condition

Firms pay a **sunk entry cost** f_e before drawing their productivity ϕ from the Pareto distribution.

The **expected profit** must equal the entry cost in equilibrium:

$$E[\pi] = \int_{\phi_d^*}^{\infty} \left[\frac{B}{\sigma} \phi^{\sigma-1} - wf_d \right] dG(\phi) + \int_{\phi_x^*}^{\infty} \left[\frac{B^*}{\sigma} \left(\frac{\phi}{\tau} \right)^{\sigma-1} - wf_x \right] dG(\phi) = wf_e$$

With Pareto distribution, this equation closes the model and determines the cutoffs.

3. Home Market Effect in Heterogeneous Firms Model

The **Home Market Effect (HME)** is stronger in the Melitz model than in the **original Krugman model**.

Setup

- Two countries: Home (larger) and Foreign (smaller)
- $L_H > L_F \rightarrow R_H > R_F$ (larger domestic expenditure in Home)
- Identical technology, fixed costs, and trade costs

Mechanism

Because $R_H > R_F$, the domestic demand index $B_H > B_F$. From the cutoff equations:

$$\phi_d^{H*} < \phi_d^{F*}$$

Results:

1. **Lower domestic cutoff in large country:** More firms can survive in the large market.
2. **Lower export cutoff for firms in large country:** Firms from the large country find it easier to export (ϕ_x^{H*} is relatively lower).
3. **Disproportionate export share:** The larger country exports a **more than proportional** share of varieties relative to its size.

Intuition: The **large domestic market** allows firms to cover fixed costs more easily. This gives firms based in the large country a cost advantage (they operate at lower average cost due to scale), making them more competitive in the export market as well.

Mathematical Expression of HME

The share of firms from Home that export is higher, and the value of exports from Home to Foreign exceeds the size ratio:

$$\frac{\text{Exports from H to F}}{\text{Exports from F to H}} > \frac{R_H}{R_F} > 1$$

This is the strong form of the Home Market Effect in the Melitz model.

Summary of Key Insights

- **Productivity cutoffs** (ϕ_d^* and ϕ_x^*) are **endogenous** and respond to market size, trade costs, and fixed costs.
- **Trade liberalization** lowers ϕ_x^* (more exporters) and raises ϕ_d^* (exit of least productive firms) → **reallocation toward more productive firms**.
- The **Home Market Effect** is **amplified**: larger countries host more firms and have a disproportionately large export share.

Definition of Aggregate Productivity

In the Melitz model, **aggregate (industry) productivity** $\tilde{\phi}$ is defined as the **weighted average productivity** that enters the price index and welfare calculations. It is given by:

$$\tilde{\phi} = \left[\frac{1}{M} \int_{\phi_d^*}^{\infty} \phi^{\sigma-1} dM(\phi) \right]^{\frac{1}{\sigma-1}}$$

where:

- M = mass (number) of producing firms; ϕ_d^* = domestic productivity cutoff
- σ = elasticity of substitution

This is the CES-weighted average productivity. Higher $\tilde{\phi}$ means lower industry price index and higher welfare.

Productivity Distribution

Firms draw productivity ϕ from a **Pareto distribution**:

$$G(\phi) = 1 - \left(\frac{\phi_{min}}{\phi} \right)^k \quad \text{for } \phi \geq \phi_{min}, \quad g(\phi) = k\phi_{min}^k \phi^{-(k+1)}$$

where $k > \sigma - 1$ (shape parameter – higher k means less dispersion). The conditional distribution of productivity among **active firms** (those with $\phi \geq \phi_d^*$) has a nice truncated Pareto form.

Derivation of Aggregate Productivity

Let M_d = number of firms producing domestically ($M_d = M$ in closed economy). The integral in the numerator is:

$$\int_{\phi_d^*}^{\infty} \phi^{\sigma-1} dG(\phi) = \int_{\phi_d^*}^{\infty} \phi^{\sigma-1} \cdot k\phi_{min}^k \phi^{-(k+1)} d\phi = k\phi_{min}^k \int_{\phi_d^*}^{\infty} \phi^{\sigma-1-k-1} d\phi = k\phi_{min}^k \int_{\phi_d^*}^{\infty} \phi^{-(k-\sigma+1)} d\phi$$

Solving the integral:

$$= k\phi_{min}^k \left[\frac{\phi^{-(k-\sigma)}}{-(k-\sigma)} \right]_{\phi_d^*}^{\infty} = \frac{k}{k-\sigma+1} \phi_{min}^k (\phi_d^*)^{-(k-\sigma)}$$

Now, the probability of surviving (being active):

$$1 - G(\phi_d^*) = \left(\frac{\phi_{min}}{\phi_d^*} \right)^k$$

So, the **average** $\phi^{\sigma-1}$ among active firms is:

$$\frac{1}{1 - G(\phi_d^*)} \int_{\phi_d^*}^{\infty} \phi^{\sigma-1} g(\phi) d\phi = \frac{k}{k - \sigma + 1} \left(\frac{\phi_{min}^k}{(\phi_d^*)^k} \right) (\phi_d^*)^{\sigma-1} \cdot (\phi_d^*)^{k-\sigma}$$

Simplifying:

$$= \frac{k}{k - \sigma + 1} (\phi_d^*)^{\sigma-1}$$

Therefore, the aggregate productivity becomes:

$$\tilde{\phi} = \left[\frac{k}{k - \sigma + 1} \right]^{\frac{1}{\sigma-1}} \phi_d^*$$

Final Clean Formula:

$$\tilde{\phi} = \left(\frac{k}{k - \sigma + 1} \right)^{\frac{1}{\sigma-1}} \phi_d^*$$

This shows that aggregate productivity $\tilde{\phi}$ is **proportional to the cutoff** ϕ_d^* .

Important Implications

- Since $\tilde{\phi} \propto \phi_d^*$, anything that **raises the domestic cutoff** ϕ_d^* increases average industry productivity.
- When trade opens:
 - ϕ_d^* **risks** (least productive firms exit)
 - $\tilde{\phi}$ **increases** → this is the **reallocation gain**

This productivity gain is **new** compared to the Krugman model and can be quantitatively large.